1- Using the continuity equation $\frac{\partial \rho}{\partial t} = -\text{div}\, \rho\langle v|$, prove:

$$\frac{\partial}{\partial t}\rho\frac{v^2}{2} + \text{div}\,\rho\frac{v^2}{2}\langle v| = \rho\frac{D}{Dt}\frac{v^2}{2}$$

$$\frac{\partial}{\partial t}\rho|v\rangle + |\,\text{div}\,\rho|v\rangle\langle v|\,\rangle = \rho\frac{D}{Dt}|v\rangle$$

a.) $\frac{\partial \rho}{\partial t} = -\text{div}\,\rho\langle v| = -\begin{bmatrix}\frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\end{bmatrix}\rho\begin{bmatrix}v_x\\ v_y\\ v_z\end{bmatrix} = -\left(\frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z)\right)$

$\frac{\partial}{\partial t}\rho\frac{v^2}{2} + \text{div}\,\rho\frac{v^2}{2}\langle v| = \rho\frac{D}{Dt}\frac{v^2}{2}$

$\frac{v^2}{2}\left(\frac{\partial}{\partial t}\rho + \nabla\cdot\rho\langle v|\right) = \frac{v^2}{2}\rho\left(\frac{D}{Dt}\right)$

$\frac{D}{Dt}\rho = \frac{\partial}{\partial t} + \nabla\cdot\langle v|$

b.) $\frac{\partial}{\partial t}\rho|v\rangle + |\,\text{div}\,\rho|v\rangle\langle v|\,\rangle$

2- Using cartesian coordinates, prove for a fluid of constant density

$$\text{div}\,|\nabla v|^\dagger = 0$$

$\text{div}[\nabla v]^\dagger = \nabla^2 v =$

$\nabla(\nabla\cdot v) - \nabla\times(\nabla\times v)$

@ constant ρ, $\nabla\cdot v = 0$

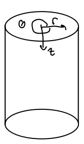
$\nabla^2 v = \nabla(0) - \nabla\times(\nabla\times v)$

and we assume irrotational flow ∴ $\nabla\times v = 0$

then

$\text{div}[\nabla v]^\dagger = -\nabla\times(\nabla\times v) = -\nabla\times(0) = 0$

2. By making a mass balance over a volume element $(\Delta r)(r\Delta\theta)(\Delta z)$ derive the equation of continuity in cylindrical coordinates.



Δr

Δz

$\frac{dm}{dt} = \frac{\partial\rho}{\partial t}(dV) = \frac{\partial\rho}{\partial t}(\Delta r\, r\Delta\theta\,\Delta z)$

$\rho = \frac{m}{V}$ $\quad dV = (\Delta r)(r\Delta\theta)(\Delta z)$

$\frac{dm}{dt} = \dot m_{in} - \dot m_{out}$

$\frac{dm}{dt} + \dot m_{out} - \dot m_{in} = 0$

$\frac{dm}{dt} + \dot m_r + \dot m_\theta + \dot m_z = 0 \Rightarrow \frac{\partial\rho}{\partial t}(r\,dr\,d\theta\,dz) + \frac{\partial}{\partial r}(\rho v_r r)\,d\theta\,dr\,dz + \frac{\partial}{\partial\theta}(\rho v_\theta)\,d\theta\,dr\,dz + \frac{\partial}{\partial z}(\rho v_z) r\,dr\,d\theta\,dz$

$$\boxed{\frac{\partial\rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(\rho v_r r) + \frac{1}{r}\frac{\partial}{\partial\theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0}$$

$m = \rho V$

$\frac{dm}{dt} = \frac{\partial}{\partial t}(\rho V)$

$\frac{dm}{dt} = \underbrace{\frac{\partial\rho}{\partial t}(dV)}_{\text{rate of accumulation}} = \frac{\partial\rho}{\partial t}(r\,dr\,d\theta\,dz)$

$\dot m_r = \rho v_r A_r = \rho v_r(r\,d\theta\,dz)$

$= (\rho v_r) r (d\theta\,dz)\Big|_r^{r+dr}$

$= (\rho v_r)_{r+dr}(r+dr)\,d\theta\,dz - ((\rho v_r)_r r(d\theta\,dz))$

$= (\rho v_r)_{r+dr}(r+dr)\,d\theta\,dz$

$= (\rho v_r r)|_r + \frac{\partial}{\partial r}(\rho v_r r)\Big|_r dr + O(dr^2) - ((\rho v_r)_r r(d\theta\,dz))$

$= \frac{\partial}{\partial r}(\rho v_r r)\,d\theta\,dr\,dz$

dr & dz because we span θ by those parameters.

$\dot m_\theta = \rho v_\theta A_\theta = \rho v_\theta(dr\,dz)$

$= (\rho v_\theta)(dr\,dz)\Big|_\theta^{\theta+d\theta} = [(\rho v_\theta)_{\theta+d\theta}(\theta+d\theta)dr\,dz] - [(\rho v_\theta)_\theta(\theta)dr\,dz]$

$= \frac{\partial}{\partial\theta}(\rho v_\theta)\,d\theta\,dr\,dz$

$\dot m_z = \rho v_z A_z = \rho v_z(dr\,d\theta)$

$= \rho v_z r\,dr\,d\theta\Big|_z^{z+dz} = (\rho v_z)_{z+dz}(z+dz)\, r\,dr\,d\theta - (\rho v_z)_z r\,dr\,d\theta$

$= \frac{\partial}{\partial z}(\rho v_z) r\,dr\,d\theta\,dz$

3B.1 Flow between coaxial cylinders and concentric spheres.

(a) The space between two coaxial cylinders is filled with an incompressible fluid at constant temperature. The radii of the inner and outer wetted surfaces are κR and R, respectively. The angular velocities of rotation of the inner and outer cylinders are Ω_i and Ω_o. Determine the velocity distribution in the fluid and the torques on the two cylinders needed to maintain the motion.

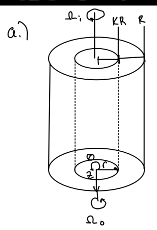
(b) Repeat part *(a)* for two concentric spheres.

Answers:

(a) $v_\theta = \dfrac{\kappa R}{1-\kappa^2}\left[(\Omega_o - \Omega_i\kappa^2)\left(\dfrac{r}{\kappa R}\right) + (\Omega_i - \Omega_o)\left(\dfrac{\kappa R}{r}\right)\right]$

(b) $v_\phi = \dfrac{\kappa R}{1-\kappa^3}\left[(\Omega_o - \Omega_i\kappa^3)\left(\dfrac{r}{\kappa R}\right) + (\Omega_i - \Omega_o)\left(\dfrac{\kappa R}{r}\right)^2\right]\sin\theta$

a.)



Assumptions:

$V_\phi(r=KR) = KR\,\Omega_i$

$V_\phi(r=R) = R\,\Omega_o$

ρ = Constant.

Volume doesn't change

@ Steady state

$V_r = 0$

$V_z = 0$

$V_\phi = V_\phi(r)$

Eqn of Motion:

$$\rho\left(\frac{V_\theta}{r}\frac{\partial V_\theta}{\partial\theta}\right) = -\frac{1}{r}\frac{\partial p}{\partial\theta} + \rho g_\theta$$

$$\rho\left(\cancel{\frac{\partial V_\theta}{\partial t}}^{0} + \cancel{V_r}^{0}\frac{dV_\theta}{dr} + \frac{V_\theta}{r}\cancel{\frac{dV_\theta}{\partial\theta}}^{\partial} + \cancel{V_z}^{0}\cancel{\frac{\partial V_\theta}{\partial z}}^{0} + \cancel{\frac{V_r V_\theta}{r}}^{0}\right) = -\frac{1}{r}\cancel{\frac{\partial p}{\partial\theta}}^{0} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rV_\theta)\right) + \frac{1}{r^2}\cancel{\frac{d^2V_\theta}{\partial\theta^2}}^{0} + \frac{2}{r^2}\cancel{\frac{\partial V_r}{\partial\theta}}^{0}\right] + \rho\cancel{g_\theta}^{0}$$

$$0 = \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rV_\theta)\right)\right]$$

$$0 = \frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rV_\theta)\right)$$

$$C_1 = \frac{1}{r}\frac{d}{\partial r}(rV_\theta)$$

$$\int C_1 r = \int\frac{\partial}{\partial r}(rV_\theta)$$

$$C_2 + \frac{1}{2}C_1 r^2 = rV_\theta$$

$$\frac{C_2}{r} + C_1 r = V_\theta(r)$$

@ $r = KR$
$V_\theta = \Omega_i$

@ $r = R$
$V_\theta = \Omega_o$

$V = r\Omega$

$\dfrac{C_2}{KR} + C_1 KR = KR\,\Omega_i$; $\dfrac{C_2}{R} + C_1 R = R\,\Omega_o$

$C_2 = (KR)^2(\Omega_i - C_1)$

$$\frac{(KR)^2\Omega_i - (KR)^2 C_1}{R} + C_1 R = R\,\Omega_o$$

$$K^2\Omega_i - K^2 C_1 + C_1 = \Omega_o$$

$$K^2\Omega_i - \Omega_o = C_1(K^2 - 1)$$

$$\frac{K^2\Omega_i - \Omega_o}{(K^2-1)} = C_1$$

$$C_2 = (KR)^2\left(\Omega_i - \frac{K^2\Omega_i - \Omega_o}{(K^2-1)}\right)$$

$$= (KR)^2\left(\frac{\cancel{\Omega_i K^2} - \Omega_i}{(K^2-1)} - \frac{\cancel{K^2\Omega_i} - \Omega_o}{(K^2-1)}\right)$$

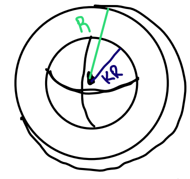
$$C_2 = (KR)^2\left(\frac{\Omega_o - \Omega_i}{K^2-1}\right)$$

$$V_\theta = r\left(\frac{K^2\Omega_i - \Omega_o}{(K^2-1)}\right) + \frac{(KR)^2}{r}\left(\frac{\Omega_o - \Omega_i}{K^2-1}\right)$$

$$V_\theta = \frac{KR}{K^2-1}\left(\frac{r}{KR}(\Omega_i K^2 - \Omega_o) + \left(\frac{KR}{r}\right)(\Omega_o - \Omega_i)\right)$$

$$\boxed{V_\theta = \frac{KR}{K^2-1}\left(\frac{r}{KR}(\Omega_o - \Omega_i K^2) + \frac{KR}{r}(\Omega_i - \Omega_o)\right)}$$

b.)



incompressible fluid at constant T

Assumptions:

$V_\phi(r=KR) = KR\,\Omega_i\sin\theta$

$V_\phi(r=R) = R\,\Omega_o\sin\theta$

ρ = Constant.

Volume doesn't change

@ Steady state

$V_r = 0$

$V_z = 0$

$V_\phi = V_\phi(r)$

$$\rho\left(\cancel{\frac{V_\phi}{\partial t}}^{0} + \cancel{V_r\frac{\partial V_\phi}{\partial r}}^{0} + \cancel{\frac{V_\theta}{r}}^{0}\frac{\partial V_\phi}{\partial\theta} + \frac{V_\phi}{r\sin\theta}\cancel{\frac{\partial V_\phi}{\partial\phi}} + \frac{V_\phi V_r + \cancel{V_\theta V_\phi\cot\theta}^{0}}{r}\right) = -\frac{1}{r\sin\theta}\cancel{\frac{\partial p}{\partial\phi}}^{0} + \mu\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V_\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial\theta}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(V_\phi\sin\theta)\right) + \cancel{\frac{1}{r^2\sin^2\theta}\frac{\partial^2 V_\phi}{\partial\phi^2}}^{0} + \cancel{\frac{2}{r^2\sin\theta}\frac{\partial V_r}{\partial\phi}}^{0} + \cancel{\frac{2\cot\theta}{r^2\sin\theta}\frac{\partial V_\theta}{\partial\phi}}^{0}\right] + \cancel{\rho g_\phi}^{0}$$

$$0 = \mu\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V_\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial\theta}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(V_\phi\sin\theta)\right)\right]$$

$$0 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V_\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial\phi}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(V_\phi\sin\theta)\right)$$

$$0 = \frac{\partial}{\partial r}\left(r^2\frac{\partial V_\phi}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(V_\phi\sin\theta)\right)$$

$$0 = \frac{d}{dr}\left(r^2\frac{df}{dr}\right) - 2f$$

$$f = C_1 r + \frac{C_2}{r^2}$$

$$V_\phi(r,\theta) = \left(C_1 r + \frac{C_2}{r^2}\right)\sin\theta$$

$KR\,\Omega_i\sin\theta = \left(C_1 KR + \dfrac{C_2}{(KR)^2}\right)\sin\theta \Rightarrow KR\,\Omega_i = C_1 KR + \dfrac{C_2}{(KR)^2} \Rightarrow KR\,\Omega_i = \Omega_o KR - \dfrac{C_2 K}{R^2} + \dfrac{C_2}{K^2R^2}$

$R\,\Omega_o\sin\theta = \left(C_1 R + \dfrac{C_2}{R^2}\right)\sin\theta \Rightarrow R\,\Omega_o = C_1 R + \dfrac{C_2}{R^2}$

$KR(\Omega_i - \Omega_o) = \dfrac{C_2}{K^2R^2} - \dfrac{C_2 K}{R^2}$

$C_1 = \Omega_o - \dfrac{C_2}{R^3}$

$KR(\Omega_i - \Omega_o) = \dfrac{C_2}{K^2R^2} - \dfrac{C_2 K^3}{K^2R^2}$

$KR(\Omega_i - \Omega_o) = C_2\dfrac{(1-K^3)}{K^2R^2}$

$C_1 = \Omega_o - \dfrac{K^3(\Omega_i - \Omega_o)}{(1-K^3)}$, $\dfrac{(KR)^3(\Omega_i - \Omega_o)}{(1-K^3)} = C_2$

$$f = r\left(\Omega_o - \frac{K^3(\Omega_i - \Omega_o)}{(1-K^3)}\right) + \frac{(KR)^3(\Omega_i - \Omega_o)}{r^2(1-K^3)}$$

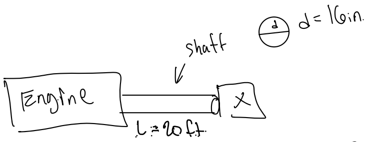
$$V_\phi = f\sin\theta$$

$$V_\phi = \frac{KR}{(1-K^3)}\left((\Omega_o - \Omega_i K^3)\left(\frac{r}{KR}\right) + (\Omega_i - \Omega_o\left(\frac{KR}{r}\right)^2\right)\sin\theta$$

3A.2 Friction loss in bearings.[1] Each of two screws on a large motor-ship is driven by a 4000-hp engine. The shaft that connects the motor and the screw is 16 in. in diameter

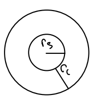
and rests in a series of sleeve bearings that give a 0.005 in. clearance. The shaft rotates at 50 rpm, the lubricant has a viscosity of 5000 cp, and there are 20 bearings, each 1 ft in length. Estimate the fraction of engine power expended in rotating the shafts in their bearings. Neglect the effect of the eccentricity.

Answer: 0.115



$\Omega_s = 50\,rpm$

$V_\theta = \Omega r$



$r_s = 8in$

Lubricant $\tau_{\theta r} = 5000\,cp$

$r_c = 0.005\,in$

$8in\left(0.0254\frac{m}{in}\right) = 0.2032m$

$C = 0.005in\left(0.0254\frac{m}{in}\right) = 1.27E\text{-}4m$

$5000\,cp = 5\,\frac{Pa}{s}$

$L = 1ft = 0.3048m$

$n = 20$

$P_{eng} = 4000hp \cdot 745W$

$\Omega = \frac{50\ \frac{rev}{min}}{60\ \frac{sec}{min}} \cdot 2\pi = 5.236\,\frac{rad}{s}$

$$5000 = -\mu\left[r\frac{\partial}{\partial r}\left(\frac{V_\theta}{r}\right) + \frac{1}{r}\left(\frac{\partial v_r}{\partial \theta}\right)\right]$$

$$\tau_{\theta r} = -\mu\left[r\frac{\partial}{\partial r}(\Omega) + \frac{1}{r}\left(\cancel{\frac{\partial v_r}{\partial \theta}}^{\,0}\right)\right] \quad \Big|\ \frac{\partial}{\partial \theta} = 0,\ v_r = 0$$

$$\tau_{\theta r} = \mu\left[r\frac{d\Omega}{dr}\right] \ \vdots\ \left.\frac{\partial V_\theta}{\partial r}\right|_{r=a} \approx \frac{V_\theta(a) - 0}{C} = \frac{a\Omega}{c}$$

$$\left.\tau_{r\theta}\right|_{r=a} \approx \frac{a\Omega}{c}$$

<u>Torque:</u>

$$T = \int_{SA} (\tau_{r\theta}\, dA)a = \tau(a)a(2\pi a L)$$

$$T = 2\pi\mu L\frac{a^3\Omega}{c}$$

$$P = T\Omega = 2\pi\mu L\frac{a^3\Omega^2}{c}$$

$$P_b = 2\pi(5)(0.3048)\left(\frac{(0.2032)^3 m^3 (5.236)^2 \frac{rad^2}{s^2}}{1.27E\text{-}4m}\right)$$

$$P_b = 17343W = 17.34\,KW \cdot \frac{1hp}{745.7\,W} = 23.26hp$$

$$P_T = nP_b\Big|^{n=20} = 20\cdot(23.26) = 465.152hp$$

$$f_w = \frac{P_b}{P_{eng}} = \frac{465.15hp}{4000hp} = \boxed{0.11629}$$

Fig. 3B.4. Creeping flow in the region between two stationary concentric spheres.

3B.4 Creeping flow between two concentric spheres (Fig. 3B.4). A very viscous Newtonian fluid flows in the space between two concentric spheres, as shown in the figure. It is desired to find the rate of flow in the system as a function of the imposed pressure difference. Neglect end effects and postulate that v_θ depends only on r and θ with the other velocity components zero.

(a) Using the equation of continuity, show that $v_\theta \sin\theta = u(r)$, where $u(r)$ is a function of r to be determined.

(b) Write the θ-component of the equation of motion for this system, assuming the flow to be slow enough that the $[\mathbf{v}\cdot\nabla\mathbf{v}]$ term is negligible. Show that this gives

$$0 = -\frac{1}{r}\frac{\partial \mathcal{P}}{\partial \theta} + \mu\left[\frac{1}{\sin\theta}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{du}{dr}\right)\right] \qquad (3B.4\text{-}1)$$

(c) Separate this into two equations

$$\sin\theta\frac{d\mathcal{P}}{d\theta} = B; \qquad \frac{\mu}{r}\frac{d}{dr}\left(r^2\frac{du}{dr}\right) = B \qquad (3B.4\text{-}2, 3)$$

where B is the separation constant, and solve the two equations to get

$$B = \frac{\mathcal{P}_2 - \mathcal{P}_1}{2\ln\ \cot\frac{1}{2}\varepsilon} \qquad (3B.4\text{-}4)$$

$$u(r) = \frac{(\mathcal{P}_1 - \mathcal{P}_2)R}{4\mu\ln\ \cot(\varepsilon/2)}\left[\left(1-\frac{r}{R}\right)+\kappa\left(1-\frac{R}{r}\right)\right] \qquad (3B.4\text{-}5)$$

where $\mathcal{P}_1$ and $\mathcal{P}_2$ are the values of the modified pressure at $\theta = \varepsilon$ and $\theta = \pi - \varepsilon$, respectively.

(d) Use the results above to get the mass rate of flow

$$w = \frac{\pi(\mathcal{P}_1 - \mathcal{P}_2)R^3(1-\kappa)^3\rho}{12\mu\ln\ \cot(\varepsilon/2)} \qquad (3B.4\text{-}6)$$

a.) $\cancel{\frac{\partial \rho}{\partial t}} + \cancel{\frac{1}{r^2}\frac{\partial}{\partial r}(\rho r^2 v_r)} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\rho v_\theta \sin\theta) + \cancel{\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}(\rho v_\phi)} = 0$

$v_r = 0$

$v_\phi = 0$

$\frac{\partial\rho}{\partial t} = 0$

$v_\theta = v_\theta(r,\theta)$

$\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\rho v_\theta \sin\theta) = 0$

$\rho v_\theta \sin\theta = C_1$

$v_\theta \sin\theta = C_1 = u(r)$

Since $\frac{d}{d\theta}(C_1) \neq 0$ C_1 wouldn't be constant. And our components for $v_\phi = \frac{\partial}{\partial\theta} = 0$ as told by the problem therefore C_1 has to be a function of r.

$v_\theta \sin\theta = u$

$v_\theta = \frac{u}{\sin\theta}$

b.) $\rho\left(\cancel{\frac{\partial v_\theta}{\partial t}} + \cancel{v_r\frac{\partial v_\theta}{\partial r}} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta} + \cancel{\frac{v_\phi}{r\sin\theta}\frac{\partial v_\theta}{\partial\phi}} + \cancel{\frac{v_r v_\theta - v_\phi^2\cot\theta}{r}}\right)$

$= -\frac{1}{r}\frac{\partial P}{\partial\theta} + \mu\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial v_\theta}{\partial r}\right) + \cancel{\frac{1}{r^2}\frac{\partial}{\partial\theta}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(v_\theta\sin\theta)\right)} + \cancel{\frac{1}{r^2\sin^2\theta}\frac{\partial^2 v_\theta}{\partial\phi^2}} + \cancel{\frac{2}{r^2}\frac{\partial v_r}{\partial\theta}} - \cancel{\frac{2\cot\theta}{r^2\sin\theta}\frac{\partial v_\phi}{\partial\phi}}\right] + \rho g_\theta$

$\rho\left(\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta}\right) = -\frac{1}{r}\frac{\partial P}{\partial\theta} + \mu\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \frac{u}{\sin\theta}}{\partial r}\right)\right] + \rho g_\theta$

$0 = -\frac{1}{r}\frac{\partial \mathcal{P}}{\partial\theta} + \mu\left[\frac{1}{r^2}\frac{1}{\sin\theta}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)\right]$

c.) $0 = -\frac{1}{r}\frac{d\mathcal{P}}{d\theta} + \mu\left[\frac{1}{r^2\sin\theta}\frac{d}{dr}\left(r^2\frac{d(v_\theta\sin\theta)}{dr}\right)\right]$

$\frac{1}{r}\frac{d\mathcal{P}}{d\theta} = \mu\frac{1}{r^2\sin\theta}\left(\frac{d}{dr}\left(r^2\frac{du}{dr}\right)\right) \quad \Big| \; r\sin\theta$

$\sin\theta\frac{d\mathcal{P}}{d\theta} = \frac{\mu}{r}\left(\frac{d}{dr}\left(r^2\frac{du}{dr}\right)\right) = B$

$B = \sin\theta\frac{d\mathcal{P}}{d\theta}$

$\int_{\varepsilon}^{\pi-\varepsilon}\frac{B}{\sin\theta}d\theta = \int_{P_1}^{P_2}\frac{d\mathcal{P}}{d\theta}d\theta$

$\ln\left|\tan\left(\frac{\pi-\varepsilon}{2}\right)\right| - \ln\left|\tan\left(\frac{\varepsilon}{2}\right)\right| = (P_2 - P_1)$

$\ln\left|\cot\left(\frac{\varepsilon}{2}\right)\right| - \ln\left|\tan\left(\frac{\varepsilon}{2}\right)\right| = (P_2 - P_1)$

$\ln\left|\frac{\cot(\frac{\varepsilon}{2})}{\tan(\frac{\varepsilon}{2})}\right| = (P_2 - P_1)$

$B\ln\left|\cot\left(\frac{\varepsilon}{2}\right)^2\right| = (P_2 - P_1)$

$B\,2\ln\left|\cot\left(\frac{\varepsilon}{2}\right)\right| = (P_2 - P_1)$

$B = \frac{(P_2 - P_1)}{2\ln|\cot(\frac{\varepsilon}{2})|}$

$\frac{\mu}{r}\left(\frac{d}{dr}\left(r^2\frac{du}{dr}\right)\right) = B$

$\frac{d}{dr}\left(r^2\frac{du}{dr}\right) = \frac{Br}{\mu}$

$r^2\frac{du}{dr} = \frac{Br^2}{2\mu} + C_1$

$\frac{du}{dr} = \frac{B}{2\mu} + \frac{C_1}{r^2}$

$u = \frac{Br}{2\mu} - \frac{C_1}{r} + C_2$

$\frac{d}{dr}\left(r^2\frac{du}{dr}\right) = \frac{(P_2 - P_1)r}{2\ln|\cot(\frac{\varepsilon}{2})|\mu}$

B.C.: $u(R) = 0$
$u(KR) = 0$

$0 = \frac{BR}{2\mu} - \frac{C_1}{R} + C_2$

$0 = \frac{BKR}{2\mu} - \frac{C_1}{KR} + C_2 \Rightarrow 0 = \frac{BKR}{2\mu} - \frac{\frac{BKR^2}{2\mu}}{KR} + C_2 \Rightarrow$

$C_2 = -\frac{B}{2\mu}(KR + R)$

$0 = \frac{BR}{2\mu} - \frac{C_1}{R} - \left(\frac{BKR}{2\mu} - \frac{C_1}{KR}\right)$

$0 = \frac{B}{2\mu}(R - KR) - C_1\left(\frac{1}{R} - \frac{1}{KR}\right)$

$0 = \text{"}(\ldots)\text{"} + C_1\left(\frac{R-KR}{KR^2}\right)$

$0 = \frac{B}{2\mu} - C_1\left(\frac{1}{KR^2}\right) = C_1 = \frac{BKR^2}{2\mu}$

$u(r) = \frac{Br}{2\mu} - \frac{C_1}{r} + C_2 = \frac{Br}{2\mu} - \frac{BKR^2}{2\mu r} - \frac{B}{2\mu}(KR + R)$

$= \frac{B}{2\mu}\left(r - \frac{KR^2}{r} - (KR + R)\right)$

$= -\frac{BR}{2\mu}\left(\frac{r}{R} - \frac{KR}{r} - (K+1)\right)$

$u(r) = \frac{BR}{2\mu}\left(\left(\frac{r}{R} - 1\right) - K\left(1 + \frac{R}{r}\right)\right) = \boxed{\frac{(P_2 - P_1)R}{4\mu\ln|\cot(\frac{\varepsilon}{2})|}\left(\left(\frac{r}{R} - 1\right) - K\left(1 + \frac{R}{r}\right)\right)}$

$$V_\phi \sin\phi = \frac{(P_2 - P_1)R}{4\mu \ln|\cot(\frac{\epsilon}{2})|}\left(\left(\frac{r}{R}-1\right) - K\left(1+\frac{R}{r}\right)\right)$$

$$V_\phi = \frac{(P_2 - P_1)R}{4\mu \sin\phi \ln|\cot(\frac{\epsilon}{2})|}\left(1 - \frac{r}{R}\right)$$

$$\dot{m} = \rho \int_A V_\phi \, dA$$

$$= \rho \int_0^{2\pi}\int_R^{KR} V_\phi (r\sin\phi)\, dr\, d\phi \Big|^{V_\phi = \frac{u(r)}{\sin\phi}}$$

$$= \rho \int_0^{2\pi}\int_R^{KR} r u(r)\, dr\, d\phi$$

$$= 2\pi\rho \int_R^{KR} r \frac{B}{\sin\phi}\left(\left(1-\frac{r}{R}\right) + K\left(1-\frac{R}{r}\right)\right) dr$$

$$\left[1-\frac{r}{R}+K\left(1-\frac{R}{r}\right)\right]$$

$$= 2\pi\rho\frac{B}{\sin\phi}\left(\int_{KR}^{R} r\left(1-\frac{r}{R}+K\left(1-\frac{R}{r}\right)\right)\right)$$

$$\int_{KR}^{R} r - \frac{r^2}{R} + Kr - R \, dr$$

$$= \left(\frac{r^2}{2} - \frac{r^3}{3R} + \frac{K}{2}r^2 - Rr\right)\Big|_{KR}^{R} = \frac{R^2}{2} - \frac{R^2}{3} + \frac{KR^2}{2} - R^2 - \left(\frac{K^2R^2}{2} - \frac{K^3R^2}{3} + \frac{K^3R^2}{2} - KR^2\right)$$

$$= 2\pi\frac{(P_2-P_1)R^3}{4\mu\ln\cot(\frac{\epsilon}{2})}\left(\frac{1}{2} - \frac{1}{3} + \frac{K}{2} - 1 - \left(\frac{K^2}{2} - \frac{K^3}{3} + \frac{K^3}{2} - K\right)\right)$$

$$= \frac{2\pi(P_2-P_1)R^3}{4\mu\ln\cot(\frac{\epsilon}{2})}\left(-\frac{5}{6} + \frac{3K}{6} - \frac{3K^2}{6} + \frac{2K^3}{6} - \frac{3K^3}{6} - \frac{6K}{6}\right)$$

$$= \frac{\pi(P_2-P_1)R^3}{2\mu\ln\cot(\frac{\epsilon}{2})}\left(\frac{1}{6}(K-1)\right)^3 = \boxed{\frac{\pi(K-1)^3(P_2-P_1)R^3}{12\mu\ln\cot(\frac{\epsilon}{2})}}$$